

Lexical Selection in Bilinguals: Do Words in the Bilingual's Two Lexicons Compete for Selection?

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In a series of picture–word interference experiments, Catalan–Spanish bilinguals named pictures in Catalan with distractor words printed either in Catalan (same-language pairs) or in Spanish (different-language pairs). Naming was facilitated when the distractor was the name of the picture in both same- [e.g., *taula–taula* (*table* in Catalan)] and different-language pairs [e.g., *taula–mesa* (*table* in Spanish)]. We also found that the facilitation effect was larger for same-language pairs and that semantically related distractors in the same- vs different-language conditions were similarly interfering. These results are interpreted within a lexical model that assumes that only words of the target language are considered for lexical selection and can, therefore, compete for selection. Converging evidence from several experiments indicates that the facilitation asymmetry for same name distractors arises because same-language distractors can activate target phonemes via nonlexical, grapheme–phoneme conversion processes. Facilitation was obtained with phonologically related distractors for both same- and different-language conditions but not for their translations. The results suggest that nonlexical phonological processes contribute to the phonological facilitation effect. © 1999 Academic Press

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Theories of language production typically assume that a crucial stage in lexical access involves the selection of the most highly activated lexical node from the set of activated nodes. Selection is required because it is assumed that, in addition to the target lexical node, other semantically related lexical nodes are also activated by the semantic system (e.g., Caramazza, 1997; Dell, 1986; Garrett, 1980; Levelt, 1989; Roelofs, 1992; Stemberger, 1985). The ease with which the target form is selected depends, among other things, on the level of activation of the competing forms. If there are highly acti-

vated competitors, target selection is more difficult, resulting in longer naming latencies and, occasionally, in erroneous responses. This process must be more complicated for bilingual speakers who are confronted with the task of selecting the right word in the right language. The latter complicating factor raises two important issues for models of lexical access: (1) Does the semantic system activate in parallel the bilingual's two lexicons? If yes, (2) do the lexical forms in the two lexicons compete for selection or are only the forms in the target lexicon considered for selection? Most theories of bilingual lexical access assume that both lexicons are activated in parallel (e.g., De Bot, 1992; Green, 1986; Poulisse & Bongaerts, 1994; but see De Bot & Schreuder, 1993). Here we address the second question through a series of picture–word interference experiments with highly proficient Catalan–Spanish bilingual speakers. Specifically, we attempt to determine whether bilingual lexical access involves *language-specific* or *language-nonspecific selection*.

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In the picture–word interference task, participants are presented with a picture (the target) and a written word (the distractor), and they are instructed to name the picture and ignore the word. Picture naming latencies vary as a function of the relation between the picture and the word (for comprehensive reviews see Glaser, 1992; MacLeod, 1991; Roelofs, 1992). Longer response times are found with semantically (categorically) related picture–word pairs (e.g., *table–chair*) than with unrelated pairs (e.g., *table–house*).¹ The Stroop effect is perhaps the best known example of semantic interference. In contrast, facilitation is observed with phonologically (usually phonologically and orthographically) related pairs (e.g., *table–tailor*), which are named faster than unrelated pairs. Facilitation is also found when the name of the picture is used as a distractor—the so-called identity effect.

There is wide agreement that the phenomenon of word interference reflects competition among lexical items at the stage of lexical selection (see e.g., Glaser & Glaser, 1989; Levelt, Schriefers, Vorberg, Meyer, Pechman, & Haviing, 1991; Roelofs, 1992; Starreveld & La Heij, 1995, 1996).² The evidence cited in support of this interpretation includes the finding that interference disappears (or is largely reduced) when picture–word stimuli are used in tasks that do not require naming responses. For instance, Schriefers, Meyer, and Levelt (1990) failed to observe interference in a “button pressing” picture recognition task (for evidence from the related Stroop task, see Melara & Mounts, 1993; Sugg & McDonald, 1994; Virzi & Egeth, 1985). Within this framework, the increase in

response latencies found with semantically related target–distractor pairs may be explained as follows. Consider the case in which the picture *table* is presented with the superimposed word *chair*. The picture is assumed to activate its corresponding semantic representation and (to a lesser extent) the semantic representations of the related concepts chair, desk, stool, cupboard, and so forth. Activation is then assumed to spread to the lexical system. As a result, the lexical nodes ‘table,’ ‘chair,’ ‘desk,’ ‘stool,’ and ‘cupboard’ are activated to varying degrees. Therefore, the lexical node ‘chair’ is activated both by the picture *table* and by the word distractor *chair*. In contrast, the lexical form of an unrelated word (*house*) receives activation only from the word distractor. As a consequence, the discrepancy in activation levels between the target response (‘table’) and the semantically related distractor (‘chair’) is smaller than that between the target response and the unrelated distractor (‘house’). From the assumption that selection difficulty is an inverse function of the similarity in activation levels of the target and distractor lexical nodes (Roelofs, 1992; Starreveld & La Heij, 1996), related distractors are expected to lead to greater interference than unrelated distractors.

The interpretation of the facilitatory effect observed with phonologically related pairs (e.g., *table–tailor*) is more complicated. The effect can be explained by at least three separate mechanisms. As already noted, phonologically related distractors are also orthographically similar to the target. It is then possible that a phonologically related distractor activates the target’s input orthographic representation (e.g., *tailor* activates ‘tailor,’ along with other lexical orthographic neighbors such as ‘table,’ ‘tablet,’ ‘taboo,’ ‘task,’ ‘sailor,’ etc.). Activation then spreads from the lexical orthographic level to the semantic and lexical phonological levels, thereby sending some activation to the target lexical node with the consequence that its selection is facilitated (see Roelofs, Meyer, & Levelt, 1996). An alternative possibility is that the activation sent by word distractors goes beyond the lexical level and reaches the level where phonemes are represented. For example,

¹ Throughout the paper we make use of the following notation: *italics* for stimuli (pictures or words), single quotation marks for lexical representations, and double quotation marks for responses.

² This is not to say that there is agreement on all aspects of the implications of this phenomenon for theories of lexical access. Thus, for example, there is disagreement on whether the effect occurs at a modality-neutral (lemmas; e.g., Levelt, Roelofs, & Meyer, 1999) or a modality-specific level of lexical representation (e.g., Caramazza & Miozzo, 1997; Starreveld & La Heij, 1996). However, this distinction is not relevant in the present discussion and is not pursued further.

tailor activates its lexical representation which in turn sends activation to its phonemes (/t/, /e/, /l/, etc.), including those shared with the target response, thereby facilitating their selection. A third possibility derives from the finding that phonological facilitation can also be produced with nonword distractors (Costa & Sebastian-Gallés, 1998; Lupker, 1982; Rayner & Posnansky, 1978; Underwood & Briggs, 1984). For example, the picture *table* is named faster when presented with a phonologically related nonword like *taibor* than with the control nonword *nupes*. The latter result can be explained by the two processes discussed above (i.e., the nonword activates orthographically similar words) but also by assuming that nonwords directly activate the target's phonological segments by means of orthography-to-phonology conversion mechanisms postulated by current reading models (e.g., Coltheart, Curtis, Aktins, & Haller, 1993; Lukatela & Turvey, 1991; Seidenberg & McClelland, 1989; Van Orden, Pennington, & Stone, 1990). The extent to which each of these mechanisms contributes to the phonological effect is at present unclear.³ The results we report below do not unambiguously resolve this issue but add important new constraints for the interpretation of the phonological facilitation effect. Finally, concerning the identity-naming condition (i.e., naming a pictured table with the superimposed word *table*), it is generally assumed that the picture names facilitate the selection of the semantic, lexical, and phonemic features needed for target production (e.g., Glaser & Glaser, 1989; Roelofs, 1992; Starreveld & La Heij, 1996).

In a number of picture-word interference studies, targets and distractors were of different languages (e.g., Ehri & Ryan, 1980; Goodman, Haith, Guttentag, & Rao, 1985; Mägiste, 1984, 1985; Smith & Kirsner, 1982; for the Stroop variant of the task see e.g., Albert & Obler, 1978; Altarriba & Mathis, 1997; Chen & Ho,

1986; Dyer, 1971; La Heij, de Bruyn, Elens, Hartsuiker, Helaha, & van Schelven, 1990; Mägiste, 1984, 1985; Preston & Lambert, 1969; Smith & Kirsner, 1982; Tzelgov, Henik & Leiser, 1990; for a review see McLeod, 1991; Smith, 1997). To illustrate, Catalan-Spanish bilingual speakers might be shown a picture with a superimposed word in Spanish and asked to name the picture in Catalan. A major finding of these studies is the replication of the semantic interference effect obtained in monolingual studies. Thus, for example, it takes longer for Catalan-Spanish speakers to name the picture *table* in Catalan ("taula") when presented with the Spanish distractor *silla* (chair) than with the Spanish unrelated word *casa* (house).

Given the interpretation of the semantic interference effect in monolingual studies discussed above, the semantic interference found with bilingual picture-word naming tasks would seem to indicate that the lexical entries of two languages compete for selection. Specifically, the results could be explained by the hypothesis schematically represented in Fig. 1A, where the example of the Catalan-Spanish pair *taula* (table)-*silla* (chair) is illustrated. The picture *table* activates its semantic representation and its associated lexical nodes in both languages ('taula' and 'mesa'). Some activation is also sent to semantically related lexical nodes in the two languages ('cadira' and 'silla'). The distractor word *silla* activates its semantic representation and its associated lexical and semantically related nodes in both languages. Thus, the lexical node 'silla' in the Spanish lexicon is highly activated. If lexical nodes in both the Spanish and the Catalan lexicons compete for selection, *silla* would interfere with the selection of the Catalan response "taula." Therefore, the finding of semantic interference is consistent with the hypothesis of *language-nonspecific selection*.

However, this is not the only possible explanation of the semantic interference effect reported in bilingual tasks. A model that assumes that only the lexical nodes in one lexicon are considered for selection can also account for the cross-language interference effects if it is assumed (as we have done above) that the bilin-

³ The explanations given here are neutral on the question of whether there is interaction between levels of representation. The results can be equally explained by interactive (e.g., Burke, MacKay, Worthley, & Wade, 1991; Dell, 1986; Stemberger, 1985) and noninteractive theories of lexical access (e.g., Roelofs, 1992).

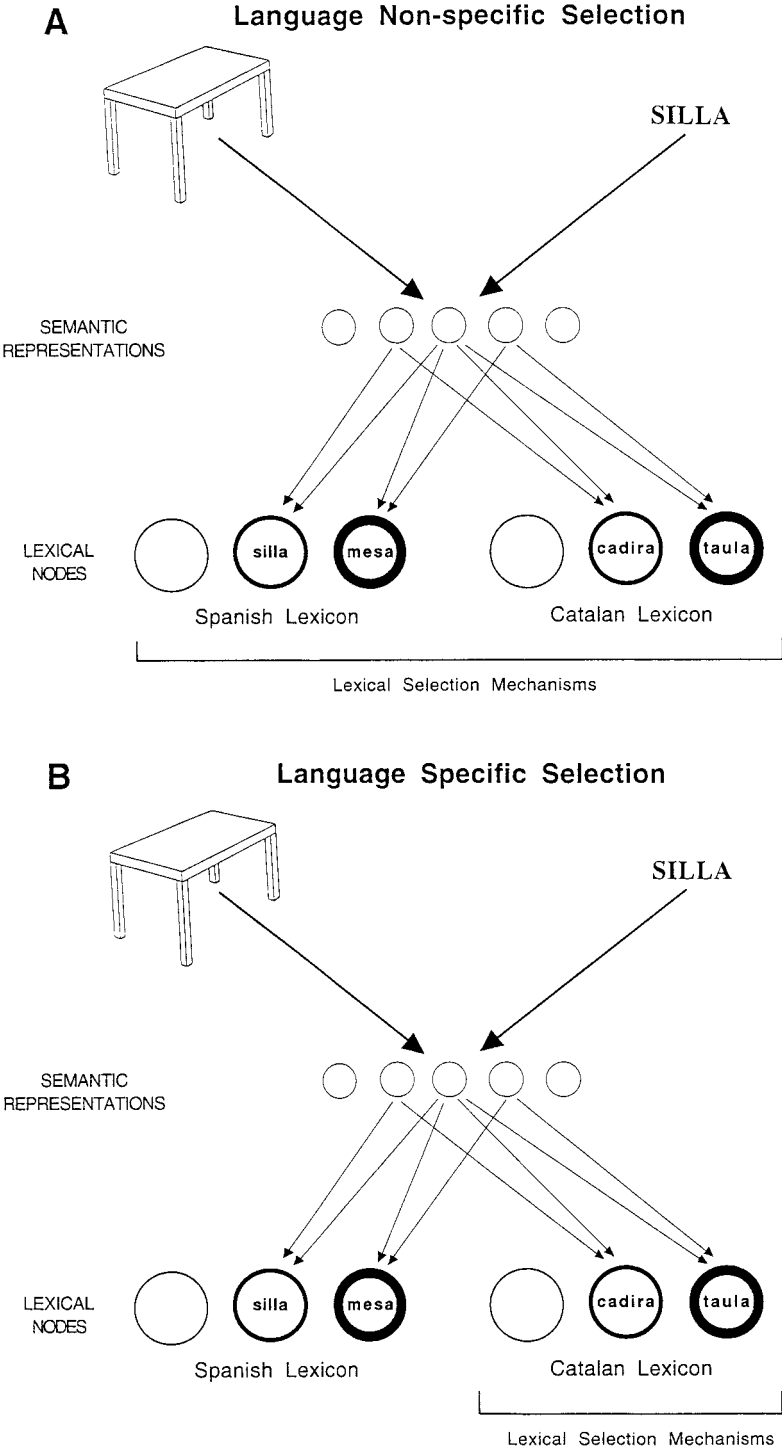


FIG. 1. Illustration of the semantic interference effect observed across languages. The Catalan–Spanish related pair *taula* (table)–*silla* (chair) is shown. (A) The hypothesis that lexical forms of the two languages (Catalan and Spanish) compete for lexical selection; (B) the hypothesis that only the forms of the predetermined language (Catalan) are considered for lexical selection. Activation is indicated by arrows. Thickness indicates the level of activation of lexical nodes.

gual's two lexicons receive activation in parallel from the semantic system. To return to the example of *taula-silla* (table-chair), the semantic representation of the Spanish word *silla* activates both the lexical node 'silla' in the Spanish lexicon and the lexical node 'cadira' in the Catalan lexicon. Additional activation spreads to the lexical nodes 'silla' and 'cadira' from the picture *table*. Thus, the Catalan lexical node 'cadira' is highly activated and would be expected to interfere with the retrieval of the Catalan word "taula" even if selection is restricted to the Catalan lexicon. That is, to explain the result of cross-language semantic interference there is no need to assume that the Spanish lexical node 'silla' interferes in the selection of the Catalan target 'taula.'

The finding of semantic interference with cross-language target-distractors does not resolve the issue of whether lexical selection is language specific or nonspecific. A more promising approach to this issue is to investigate the effect of another type of word distractor: the name of the picture presented in the nonresponse language (*different-language identity condition*). An example of this type of stimuli is given by the Catalan-Spanish pair *taula* (table)-*mesa* (table). Contrasting predictions are made by the language-specific and language-nonspecific selection hypotheses. If lexical selection is not language specific the highly activated Spanish lexical node 'mesa' would interfere with the selection of the target 'taula' in the Catalan lexicon. The lexical node 'mesa' is highly activated because it receives activation from the picture *table* and the written stimulus *mesa* (see Fig. 2A). On this account, one might even expect to find that in cross-language tasks identical distractors interfere more than semantically related distractors. In fact, if an identical picture activates the distractor's meaning more than a semantically related picture and if this difference translates into a difference in the activation of the distractor's lexical form, larger interference might be observed with identical than semantically related distractors.

Alternatively, if lexical selection is language specific, then different-language identity distractors should lead to facilitation. This expect-

tation is based on the reasoning that the Spanish distractor *mesa* (through its semantic representation) activates the lexical form of its Catalan translation 'taula' (table) and therefore further activates the target response. Because only the lexical nodes in the Catalan lexicon are considered for selection, the extra activation of 'taula' facilitates its production (see Fig. 2B).

Cross-language identity has not been investigated in cross-language picture-word interference experiments except in the study of Goodman et al. (1985). However, this study does not allow strong conclusions since contrasting results were obtained across experiments and children were tested. Thus, the issue of whether facilitation or inhibition is found with different-language identity distractors remains unresolved. This issue was investigated in the present series of experiments. The results provide useful information for determining whether lexical selection is language specific or nonspecific.

EXPERIMENT 1: DIFFERENT-LANGUAGE IDENTICAL DISTRACTORS

In Experiment 1, Catalan-Spanish speakers were asked to name a set of pictures in Catalan. Pictures were presented with a superimposed word, which could be written either in Catalan (*same-language* distractor) or in Spanish (*different-language* distractor). (Brief descriptions of the Catalan and Spanish languages are given in Appendix 1.) Two types of Catalan and Spanish distractors were used: picture names (*identical condition*) and unrelated distractors (*control condition*). To illustrate, the picture *table* (*taula* in Catalan) appeared with the following word distractors: *taula*, *mesa* (Spanish, table), *pernil* (Catalan, ham), and *jamón* (Spanish, ham).

Method

Participants. The participants in this and the following experiments were Catalan-Spanish bilingual speakers (a description of the Catalan-Spanish bilingual community is presented in Appendix 1). Participants were students at the University of Barcelona (ages 18-25) and received course credit for their participation.

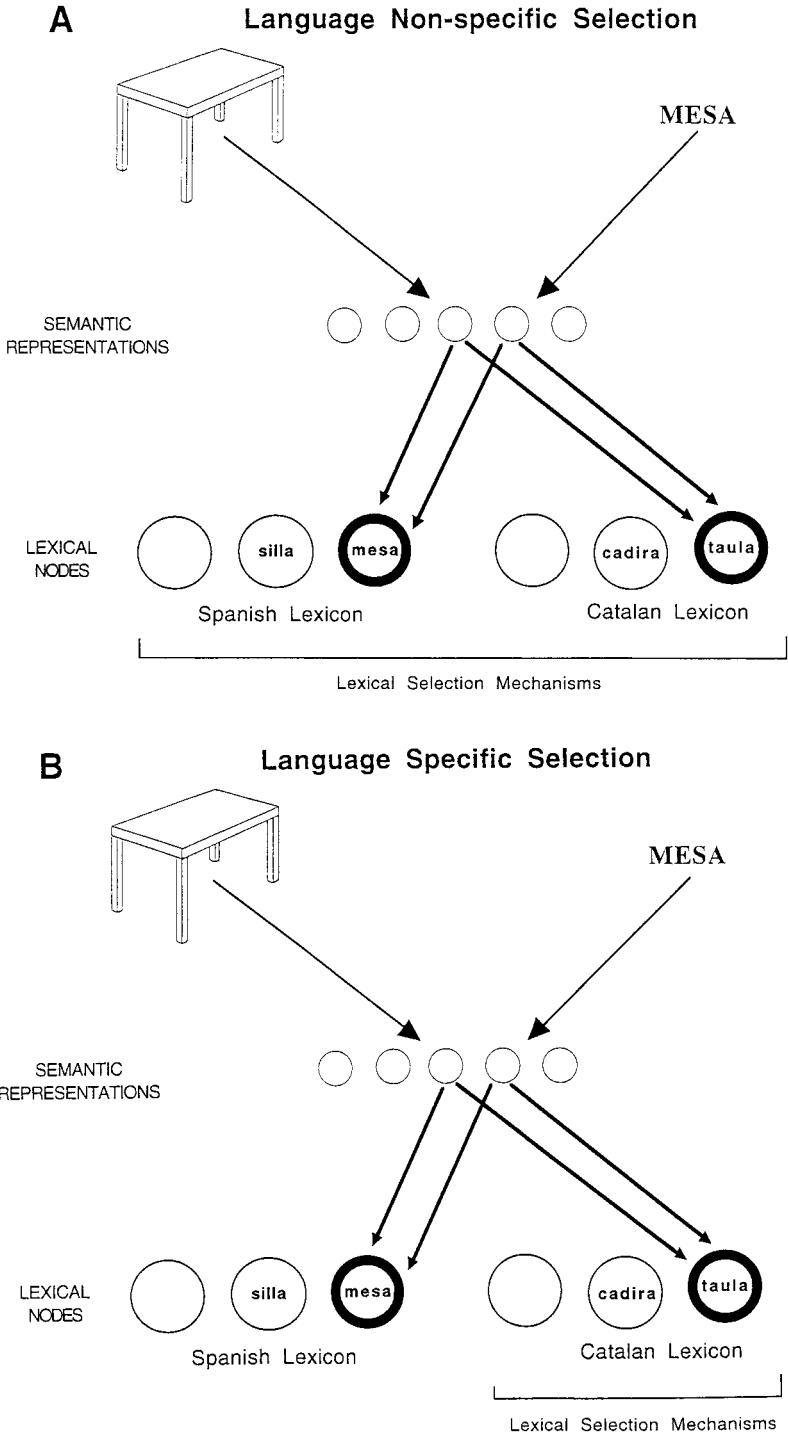


FIG. 2. Schematic representation of the effect of cross-language identical distractors. The Catalan-Spanish pair *taula* (table)-*mesa* (table) is illustrated. If forms of two languages compete for lexical selection, evidence of interference is expected with these pairs (A). If only the forms of one language are considered for lexical selection, facilitation should be observed (B). Activation is indicated by arrows. Thickness indicates the level of activation of lexical nodes.

None of the participants took part in more than one experiment. In Experiment 1 there were 18 participants.

Materials. We selected 18 pictures with noncognate names in Catalan and Spanish [i.e., they were phonologically and orthographically dissimilar; e.g., *taula*–*mesa* (table), *poma*–*manzana* (apple)]. Pictures were paired with four distractors—two Catalan nouns and their Spanish translations (the stimuli are listed in Appendix 2a). One word was the name of the picture, the other an unrelated name. Paired target–distractor nouns always had a different onset and did not rhyme. Distractors were noncognate words that were controlled both for frequency and word length (in number of letters). We report only the frequency of Spanish distractors (Sebastian, Marti, Cueto & Carreiras, 1996) since in Catalan frequency norms are only available for a limited corpus of words. Identical and unrelated Spanish distractors were similar in frequency [mean: identical = 488, unrelated = 385; $F(1,36) < 1$] and word length (mean number of letters = 6, range = 3–10; all F s were not significant). Care was taken to exclude distractors that have language-specific orthographic marking (e.g., $\ç$ or $\tilde{n}$; only two words had one of these markings). We included 144 filler trials in which the set of pictures described above along with a new set of 18 pictures appeared with unrelated word distractors. In filler trials, Catalan and Spanish distractors were equally represented. Fillers were used as warm-up stimuli in the first three trials of each block. In the training block, all the pictures were presented once and were paired either with a Catalan or a Spanish unrelated distractor. The words were shown in capital letters (Helvetica font, bold, 27 point) and were superimposed on the pictures. Pictures appeared in the center of the screen. To prevent participants from anticipating a distractor's position, word position varied randomly in the region around fixation. For a given picture, however, the distractors always appeared in the same location. Stimuli were presented in 6 blocks of 36 trials. Each picture appeared once per block. In each

block, stimuli of the various conditions appeared an equal number of times. Block trials were randomized with the restriction that distractors of the same experimental condition or of the same language appeared in no more than two consecutive trials. The order of block presentation varied across participants.

Procedure. Participants were tested individually in a soundproof booth. Instructions were administered in Catalan. Participants were instructed to name the pictures as fast and as accurately as possible in Catalan. They were informed that they would see picture–word pairs and were asked to ignore the words. Before the experiment proper, participants were presented with the entire set of pictures along with their expected Catalan names. Next, the experimenter showed all the pictures without their names and participants named the stimuli. Then, participants performed a training block of 44 trials, followed by the experiment proper. Each trial had the following structure. First, a fixation point (an asterisk) was shown in the center of the screen for 1 s, followed by a blank interval of 500 ms. Then the picture and the word were presented for 400 ms. If a response was not provided within 1.9 s, the next trial started automatically. The intertrial interval was 1.5 s. Response latencies were measured from the onset of the stimulus to the beginning of the naming response. Stimulus presentation was controlled by the Expe program (Pallier, Dupoux & Jeannin, 1997). Response latencies were measured by means of a voice key. The session lasted approximately 30 min.

Analyses. Three types of responses were scored as errors (a) production of names that differed from those designated by the experimenter; (b) verbal disfluencies (stuttering, utterance repairs, production of nonverbal sounds which triggered the voice key); and (c) recording failures. Erroneous responses and outliers (i.e., responses exceeding 1.9 s) were excluded from the analyses of response latencies. Separate analyses were carried out with subjects and items as dependent variables, yielding $F1$ and $F2$ statistics, respectively. If not otherwise stated, all the analyses should be considered as

TABLE 1

Mean Reaction Time (RT), Standard Deviation (SD), and Error Percentage for Experiment 1

Distractor	Language					
	Catalan			Spanish		
	RT	SD	Error %	RT	SD	Error %
Identical	628	78	1.8	729	78	2.0
Unrelated	766	107	2.1	754	93	1.9
Identity effect (unrelated–identical)	+138			+25		

within subject. The results of the error analyses are reported only if significant.

Results and Discussion

Table 1 shows the mean response latencies and error rates as a function of type of distractor (identical vs unrelated) and distractor language (Catalan vs Spanish). Erroneous responses were observed on 1.9% of the trials. Two variables were examined: Type of Distractor (identical vs unrelated) and Target–Distractor Language (same vs different).

Faster responses were observed with identical distractors [$F(1,17) = 44.6$, $MS_e = 2688.4$, $p < .001$; $F(1,17) = 167.1$, $MS_e = 706.3$, $p < .001$] and with same-language pairs [$F(1,17) = 72.9$, $MS_e = 481.6$, $p < .001$; $F(2,17) = 11.1$, $MS_e = 3253.5$, $p < .001$]. The interaction was significant [$F(1,17) = 46.3$, $MS_e = 1238.2$, $p < .001$; $F(2,17) = 64.5$, $MS_e = 894.1$, $p < .001$], reflecting a larger identity effect for same- than different-language pairs. A post hoc comparison revealed a significant difference between identical and unrelated stimuli in the different-language condition [$t(17) = 2.4$, $p < .02$; $t(17) = 2.8$, $p < .01$].

The facilitation in the different-language identity condition has major implications for theories of lexical access in bilingual speakers. This finding is problematic for the hypothesis that lexical nodes in the two languages of a bilingual compete for selection. As described in the Introduction (see Fig. 2A), this hypothesis predicts inhibition with pairs like *taula* (Cata-

lan, table)–*mesa* (Spanish, table). The fact that facilitation was observed is clearly incompatible with this hypothesis but is readily explained by models that postulate language-specific selection processes. A critical feature of the latter models is the assumption that the semantic system sends activation in parallel to lexical entries in both lexicons. Thus, for example if the Spanish word *mesa* (table) is presented, both the lexical node ‘*mesa*’ and its Catalan counterpart ‘*taula*’ (table) are activated. Therefore, since the Spanish word *mesa* does not interfere with the selection of the Catalan target and since *mesa* activates its Catalan twin ‘*taula*,’ it should be relatively easier to select ‘*taula*’ in the context of *mesa* than some other word. This prediction was confirmed by the results of Experiment 1. In conclusion, the finding of facilitation for the different-language identical distractor condition favors the language-specific hypothesis of lexical selection in bilingual speakers.

This conclusion has implications for the interpretation of the semantic interference effect observed with cross-language pairs like *taula* (Catalan, table)–*silla* (Spanish, chair). This effect can be explained in terms of interference occurring within one lexicon (as shown in Fig. 1B). In this example, the lexical node ‘*cadira*’ receives activation both from the semantic representation of the picture *table* and the Spanish distractor *silla*. Since only the Catalan lexicon is inspected for selection, ‘*cadira*’ (but not ‘*silla*’) interferes with the selection of the Catalan target ‘*taula*.’ There will be more interference than

TABLE 2
Examples of Target–Distractor Pairs for Experiment 2

Catalan name of the picture	Distractor			
	Language	Identical	Semantically related	Unrelated
<i>Taula</i> (table)	Catalan Spanish	Taula (table) Mesa (table)	Armari (closet) Armario (closet)	Palau (palace) Palacio (palace)

with unrelated distractors because the latter words do not receive activation from the picture.

Another result observed in Experiment 1 is that the size of the identity effect was larger for same language distractors than for different-language distractors. Detailed consideration of this result is deferred until Experiment 3. For now, we examine an alternative explanation for the finding of facilitation with cross-language identical distractors (e.g., *taula–mesa*). It could be argued that we failed to find an inhibitory effect in the different-language identity condition because participants were required to respond always in one language. The *blocked* task might have allowed participants to “focus” their lexical search on one language and minimize the interfering effect of competing lexical nodes in the nonresponse lexicon (for a similar argument see Tzelgov et al., 1990). To assess whether the results obtained in Experiment 1 were due to the use of a blocked naming task, in Experiment 2 we compared the performance of two groups of speakers. One group named pictures only in Catalan (blocked naming), and the other named pictures both in Catalan and Spanish (mixed naming). If the absence of a cross-language inhibition effect were an artifact of the use of a blocked task, it should not be reproduced with a mixed naming task.

EXPERIMENT 2: CATALAN-SPANISH MIXED NAMING RESPONSES

The objective of Experiment 2 was to determine whether the cross-language identity effect can be replicated when both languages (Catalan and Spanish) are used for response. This mixed task presumably maximizes the opportunity of

finding interference across languages. This is because both lexicons are activated and used in the course of the experiment. In this experiment, participants were required to select lexical representations in the two languages depending on a cue. In this experimental context it may not be possible to restrict selection to only one set of lexical nodes. To further test the extent to which the mixed naming task may affect the interference produced by the different-language distractors, we also included semantically related distractors. If the interference produced by different-language distractors increases when participants have to name the pictures in the two languages, we should expect more semantic interference in the mixed- as opposed to the same-language naming task, at least for different-language semantic related distractors.

Method

Participants. Two groups of 18 Catalan–Spanish speakers took part.

Material. Twenty-two pictures with Catalan and Spanish noncognate names were selected. Each picture was paired with six distractor words: three Catalan words and their Spanish translations (see Appendix 2b). The Catalan and Spanish distractors paired with a given picture were of three types: the picture name, a semantically (categorically) related word, and an unrelated word (see examples in Table 2). The Spanish translations used as semantically related and unrelated distractors were cognates of the Catalan distractors [i.e., they were orthographically and phonologically similar; e.g., *armari–armario* (closet), *palau–palacio* (palace)]. Paired target–distractor nouns always had a different onset and did not rhyme. Distractors were

TABLE 3
Mean Reaction Time (RT), Standard Deviation (SD), and Error Percentage for Experiment 2

Distractor	Language					
	Catalan			Spanish		
	RT	SD	Error %	RT	SD	Error %
Blocked Naming Task						
Identical	662	56	0.6	749	67	2.3
Semantically related	795	64	5.4	809	82	3.8
Unrelated	775	68	2.1	777	55	2.5
Identity effect (unrelated–identical)	+113			+28		
Semantic effect (unrelated–semantically related)	–20			–32		
Mixed Naming Task						
Identical	821	118	3.5	855	100	5.3
Semantically related	938	79	6.2	921	90	7.0
Unrelated	907	102	5.5	890	99	4.9
Identity effect (unrelated–identical)	+86			+35		
Semantic effect (unrelated–semantically related)	–31			–31		

controlled both for frequency and word length (in number of letters). The three sets of Spanish distractors (identical, related, and unrelated) were of comparable frequency [mean: identical = 26, related = 15, unrelated = 11; $F(2,42) = 2.4$, $MS_e = 11887$, n.s.]. Distractors in the identical, related, and unrelated conditions were of similar length in both languages (mean of number of letters = 5.8, range = 2–11; all F s were not significant). Only one distractor [*braç* (Catalan, *arm*)] had language-specific orthographic markings. A set of 22 pictures was used as fillers. Twenty filler pictures appeared four times; two filler pictures appeared six times. Half of the words paired with filler pictures were unrelated Catalan words, the other half were their Spanish translations. Stimuli were presented in four blocks of 53 trials.

Participants named the experimental pictures in Catalan. In the mixed-language naming task, participants named pictures in Spanish in 30% of the trials. For trial randomization, a new constraint was added to those of Experiment 1: responses were given in the same language in no more than three consecutive trials. The entire

experimental session lasted approximately 30 min.

Procedure. The procedure of the blocked-language naming task was identical to that of Experiment 1. In the mixed-language naming task we introduced the following changes (a) at the beginning of the experiment, the expected picture names were shown both in Catalan and Spanish; (b) a colored (red or blue) dot was used as fixation point to cue the response language; and (c) the interval between the fixation point and picture presentation was 300 ms.

Results

Table 3 shows the distribution of mean response latencies and error rates as a function of naming task (blocked- vs mixed-language), type of distractor (identical, related, unrelated), and distractor language (Catalan vs Spanish). Erroneous responses were observed on 3.9% of the trials. Separate analyses were carried out for determining whether identity and semantic effects were obtained for each task. These effects were also compared between tasks.

Identity effect. Responses were faster for identical than unrelated distractors [blocked-language naming task: $F(1,17) = 99.2$, $MS_e = 920.9$, $p < .0001$; $F(2,1,21) = 89.1$, $MS_e = 1246.9$, $p < .0001$; mixed-language naming task: $F(1,17) = 17.7$, $MS_e = 3796.6$, $p < .0001$; $F(2,1,21) = 23.6$, $MS_e = 3730.7$, $p < .0001$]. Furthermore, in the blocked-language naming task responses were faster for same-language as opposed to different-language pairs [$F(1,17) = 36.5$, $MS_e = 965.6$, $p < .0001$; $F(2,1,21) = 42.6$, $MS_e = 1046.8$, $p < .0001$]; in the mixed-language naming task, this difference was not significant [$F(1,17) < 1$; $F(2,1,21) = 1.0$, $MS_e = 1703.3$, n.s.]. Finally, in both tasks there was a significant interaction between the variables Type of Distractor and Target–Distractor Language [blocked-language naming task: $F(1,17) = 26.9$, $MS_e = 1225.0$, $p < .0001$; $F(2,1,21) = 24.5$, $MS_e = 1717.4$, $p < .0001$; mixed-language naming task: $F(1,17) = 9.1$, $MS_e = 1274.5$, $p < .01$; $F(2,1,21) = 6.1$, $MS_e = 2424.6$, $p < .02$]. This interaction shows that the identity effect was larger for same-language vs different-language pairs. A post hoc comparison revealed a significant difference between identical and unrelated stimuli in the different-language condition in both the blocked-language naming task [$t(17) = 2.9$, $p < .01$; $t(21) = 2.4$, $p = .02$] and the mixed-language naming task [$t(17) = 2.3$, $p = .03$; $t(21) = 2.8$, $p < .01$].

Semantic effect. Responses were faster for unrelated than related distractors [blocked-language naming task: $F(1,17) = 8.1$, $MS_e = 1423.4$, $p < .01$; $F(2,1,21) = 5.0$, $MS_e = 3131.8$, $p < .05$; mixed-language naming task: $F(1,17) = 8.1$, $MS_e = 2034.4$, $p < .02$; $F(2,1,21) = 4.6$, $MS_e = 4915.4$, $p < .05$]. We did not observe a significant effect of Target–Distractor Language, nor an interaction between this variable and Type of Distractor ($F_s \leq 1$).

Errors were more numerous for the related than the unrelated distractor condition [blocked-language naming task: $F(1,17) = 6.4$, $MS_e = .9$, $p < .02$; $F(2,1,21) = 4.9$, $MS_e = 1.0$, $p < .05$; mixed-language naming task: $F(1,17) = 9.6$, $MS_e = 0.7$, $p < .01$; but $F(2,1,21) = 2.8$, $MS_e = 1.5$, $p = .1$].

Between-task analysis. Separate three-way ANOVAs were carried out for each effect. In these analyses, Distractor Language and Type of Distractor were within-subject variables, and Experiment was a between-subject variable. As suggested by the null results of the triple interactions, the identity effect [$F(1,34) = 2.1$, $MS_e = 1254.6$, n.s.; $F(2,34) = 1.4$, $MS_e = 2071.0$, n.s.] and the semantic effect ($F_s < 1$) did not differ between the two tasks.

Discussion

Same-name distractors had the same effects in the two naming tasks (blocked vs mixed language). Of particular interest here is that the cross-language identity effect was replicated in the mixed-language naming task. This result demonstrates that the cross-language identity effect is a robust phenomenon that can be replicated in different conditions. Further demonstration that the mixed-language naming task did not affect the performance of subjects in the critical conditions tested in the experiment is that the semantic interference produced by different-language distractors was the same for the blocked and mixed conditions (32 and 31 ms). Therefore, even when maximizing the opportunity for competition between languages, it seems that balanced bilinguals can restrict lexical access to one of their two lexicons.

As in Experiment 1, we found a larger identity effect for same- as opposed to different-language distractors. That is, the production of the Catalan word [e.g., “taula” (table)] was facilitated more by its Catalan name (*taula*) than by its Spanish name (*mesa*). The basis for this asymmetry is explored in the remaining experiments in this article.

EXPERIMENT 3: NONCOGNATE SEMANTICALLY RELATED DISTRACTORS

Why is facilitation greater when the name of the picture is presented in the response language? A possible explanation for this result is that distractors in the response language activate their lexical nodes more than distractors in the nonresponse language. This explanation does not agree with our hypothesis that word distrac-

tors activate their lexical nodes in both lexicons equally in proficient bilinguals. To evaluate this hypothesis, we examined the effect of semantically related distractors. If same-language distractors activate their lexical nodes more than different-language distractors, we should find a larger effect of semantic interference with distractors shown in the response language. That is, the retrieval of the Catalan target “taula” (table) is expected to be hindered more by the Catalan distractor *cadira* (chair) than by its Spanish translation *silla*. The results of Experiment 2 seem to contradict this prediction—the size of the semantic interference effect was essentially identical with same- and different-language distractors. However, this result must be interpreted cautiously since it may reflect the use of cognates as distractors. Cognate words have the same meaning and almost identical forms in the two languages, as shown by the Catalan–Spanish pair *armari*–*armario* (closet). It is well known that a written word activates its own orthographic representation and that of its orthographically similar neighbors (e.g., Andrews, 1997; Segui & Grainger, 1990; Snodgrass & Mintzer, 1993). Furthermore, as demonstrated by Bijeljac-Babic, Biardeau, and Grainger (1997), neighborhood effects can be observed across languages. Based on these observations, one could argue that in our experiments (a) a distractor word activated both its own orthographic representation and that of its cognate and (b) both orthographic representations activated the distractor’s semantic representation. Thus, since both the distractor and its cognate contributed to the interference effect, it is not surprising that Catalan and Spanish distractors are equally interfering. Furthermore, the viewing conditions in the picture–word interference task are not optimal for word recognition: words are presented for limited time, embedded in a picture, and in the context of another task (picture naming). Thus, it is even possible that the distractor word was occasionally confused with its cognate. In light of these considerations, we used noncognates distractors in Experiment 3 to test whether the effect of semantic interference varies as a function of distractor language.

TABLE 4
Examples of Target–Distractor Pairs for Experiment 3

Catalan name of the picture	Distractor		
	Language	Semantically related	Unrelated
<i>Colom</i> (dove)	Catalan	Mussol (owl)	Genoll (knee)
	Spanish	Buho (owl)	Rodilla (knee)

Method

Participants. Sixteen Catalan–Spanish bilingual speakers took part in Experiment 3.

Materials and procedure. We selected 19 pictures with noncognate names in Spanish and Catalan. Each picture was paired with four noncognate word distractors: two words were in Catalan, and the other two words were their Spanish translations. Two of the words paired with a given picture were semantically (categorically) related to the target, the other two words were unrelated (see examples in Table 4; the list of stimuli is presented in Appendix 2c). Paired target–distractor words did not have the identical onset, nor did they rhyme. Related and unrelated Spanish distractors were equated for frequency [mean: related = 12, unrelated = 19; $t(18) = 1.2$, n.s.]. Related and unrelated distractors of both languages were also controlled for length (mean number of letters = 6.1, range = 3–10; all *F*s were not significant). An additional 19 pictures served as fillers. Each filler picture was paired with four unrelated distractors (two Catalan words and their Spanish translations). In the training block, pictures were shown with a superimposed string of Xs. The Xs appeared in the same position as word distractors. The stimuli were distributed across four blocks of 38 trials. Experimental conditions were counterbalanced across blocks. In all other respects, the procedure was identical to that of Experiment 1.

Results and Discussion

Table 5 shows the mean reaction times and error rates as a function of the distractor’s language (Catalan vs Spanish) and the distractor’s

TABLE 5

Mean Reaction Time (RT), Standard Deviation (SD), and Error Percentage for Experiment 3

Distractor	Language					
	Catalan			Spanish		
	RT	SD	Error %	RT	SD	Error %
Semantically related	824	97	2.8	804	97	1.8
Unrelated	792	78	3.8	782	85	3.5
<i>Semantic effect</i> (unrelated–semantically related)	–32			–22		

relatedness to the target (related vs unrelated). Errors accounted for 2.8% of the responses. Both the effects on Distractor Language and Target–Distractor Relation were examined. Responses tended to be faster with Spanish than Catalan distractors (784 vs 814 ms). This difference was significant in the subject analysis [$F(1,15) = 6.3$, $MS_e = 590.6$, $p < .05$] but not in the item analysis [$F(1,18) = 2.6$, $MS_e = 1558.9$, n.s.]. Response latencies were also faster with unrelated than related distractors [$F(1,15) = 5.9$, $MS_e = 1886.8$, $p < .03$; $F(1,18) = 6.6$, $MS_e = 2018.0$, $p < .02$], a result indicating semantic interference. Most importantly in the present context, no interaction was found between these two variables ($F_s < 1$).

The results show that the language of the distractor does not affect the magnitude of semantic interference. Other investigators have similarly failed to observe an effect of distractor language (Caramazza & Brones, 1980; Ehri & Ryan, 1980, Experiments 1 and 2; Goodman et al., 1985; La Heij, Hooglander, Kerling, & van der Valden, 1996; Mägiste, 1984, 1985; Smith & Kisner, 1982; Tzelgov et al., 1990). Yet other studies have reported an asymmetry between the semantic effects obtained in the two languages (e.g., Altarriba & Mathis, 1997; Chen & Ho, 1986; Dyer, 1971; Ehri & Ryan, 1980, Experiment 3; Preston & Lambert, 1969; Tzelgov et al., 1990). Several investigators have proposed that such discrepancies reflect the participants' level of bilingual proficiency (Kroll & Stewart, 1994; Mägiste, 1984; Potter, So, Von

Eckhardt, & Feldman, 1984). In studies where highly proficient bilingual speakers were tested, the size of the semantic interference effect is typically the same for the two languages. Our results, which were obtained with highly proficient bilinguals, fit well with this account.

The results of Experiment 3 indicate that in proficient bilinguals the lexical nodes in the response lexicon are activated to equal degrees regardless of the language in which the distractor is presented. These results are consistent with the language-specific selection hypothesis. A critical feature of this hypothesis is "automatic translation": A word distractor is assumed to activate its output lexical representations in the two languages of the bilingual speaker (see Figs. 1B and 2B). Thus, for example, the Spanish word *silla* (chair) activates the lexical node 'silla' and its Catalan translation 'cadira.' This hypothesis also assumes that the lexical nodes in the two languages are activated to the same degree. The finding that the language of the distractor does not affect the magnitude of semantic interference is consistent with our hypothesis that the activation of the lexical nodes in the response lexicon does not depend on the distractor's language.

EXPERIMENT 4: THE TIME COURSE OF SAME- AND DIFFERENT-LANGUAGE IDENTITY EFFECTS

Thus far we have examined two major results: (a) the finding of cross-language identity facilitation and (b) the finding that Catalan and Spanish semantically related distractors inter-

ferred equally. We have argued that these results are consistent with a model of lexical access that makes the following critical assumptions: (a) the semantic system activates its corresponding forms in two lexicons *equally* (e.g., the semantic representation of *table* activates both the Catalan lexical node 'taula' and the Spanish lexical node 'mesa') and (b) *only* the activated forms of the lexicon that are programmed for response are considered for selection. This type of model is schematically represented in Figs. 1B and 2B. In Experiments 1 and 2 we found that the identity effect was *larger* for same-language than different-language pairs. Thus, for example, the retrieval of the Catalan target name "taula" was facilitated more by the word distractor *taula* than by its Spanish translation *mesa*. At first glance, this asymmetry seems to challenge a model which assumes that translated forms are equally activated and that only the lexical nodes of one lexicon compete for selection. However, we argue that when other aspects of the lexical access process and the picture-word interference task are considered, the model can account for these seemingly contradictory data.

As shown in several monolingual studies of the word interference effect, it is easier to name a picture paired with a phonologically related distractor (e.g., *table-tailor*) than with an unrelated distractor (e.g., *table-sailor*; see e.g., Briggs & Underwood, 1982; Lupker, 1982; Posnansky & Rayner, 1977, 1978; Rayner & Posnansky, 1978; Rayner & Springer, 1986; Starreveld & La Heij, 1995; Underwood & Briggs, 1984). Importantly, this effect of phonological facilitation has also been obtained with phonologically related *nonwords* (as in *table-taibor*). The observation of a phonological effect with nonwords suggests that these distractors can activate the phonological content of the target word by means of nonlexical processes. Whether these processes involve the grapheme-to-phoneme (sublexical) conversions

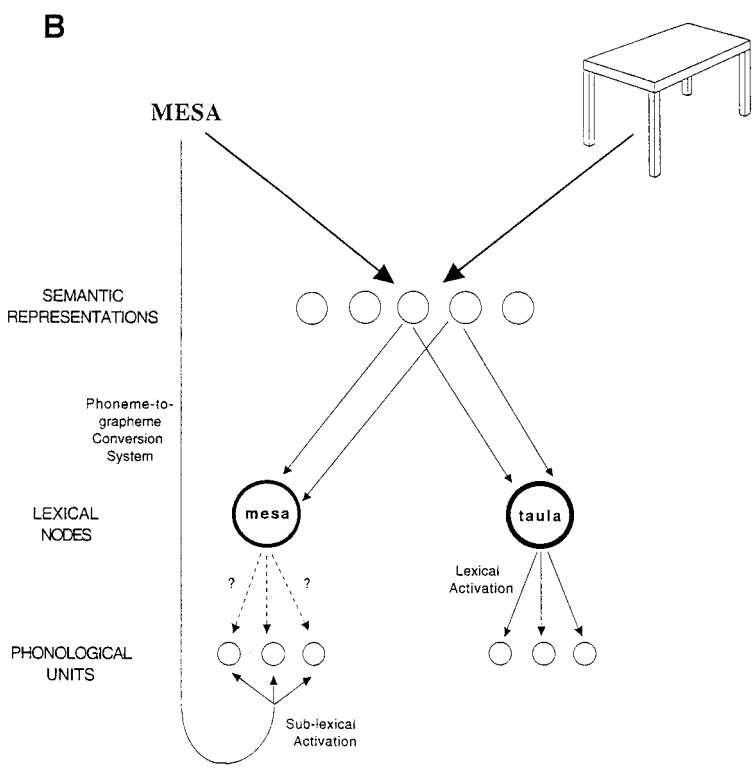
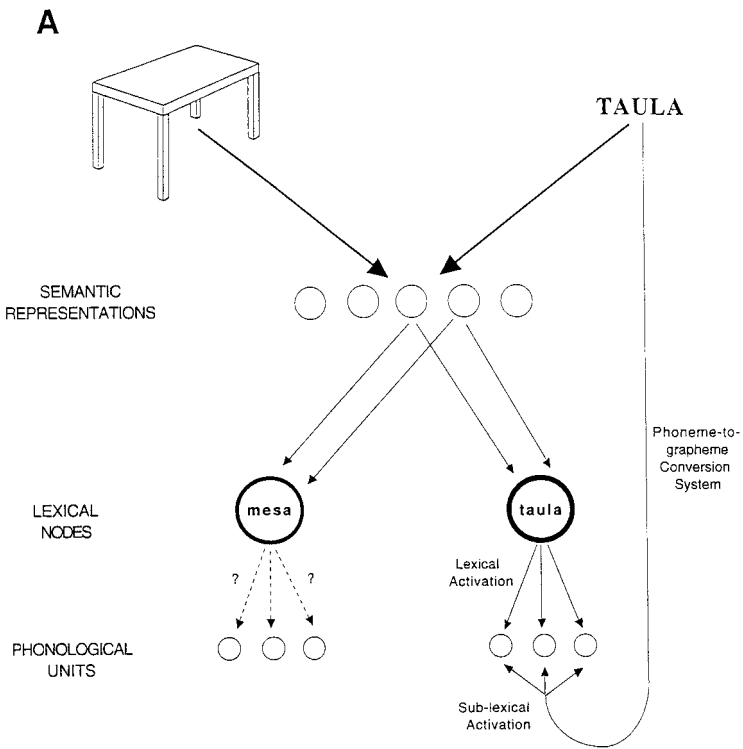
mechanisms hypothesized by dual-route models of reading (e.g., Coltheart et al., 1993) or whether the mechanisms are of some other nature (e.g., Lukatela & Turvey, 1991; Seidenberg & McClelland, 1989; Van Orden et al., 1990) is irrelevant here. What is crucial in this context is that these nonlexical processes might also come into play with word distractors. That is, part of the phonological effect observed with words could be due to activation from sublexical processing of the word distractors.⁴

The latter claim provides the basis for explaining the asymmetry in the identity effects that was reported in Experiments 1 and 2. In the same-language identity condition, target and distractor have identical forms. Thus, the word distractor can activate "nonlexically" all the segments of the target name. This activation facilitates the selection of the target's phonological segments (see Fig. 3A). In contrast, the only manner in which different-language identical distractors can facilitate target naming is by lexical mediation—the words *mesa* and *taula* do not share many segments (see Fig. 3B). Thus, in the same-language identity condition two facilitatory effects are at play: one *lexical*, the other *phonological*; in contrast, in the different-language identity condition there is only a lexical source of facilitation. This asymmetry can account for the greater facilitation observed in the same-language condition.

Experiment 4 was designed to provide empirical support for the hypothesis that different levels of processing are implicated in the same- and different-language identity effects. To this end, we

⁴ In the Introduction we mentioned an alternative explanation of the phonological effect. Namely, phonologically related distractors activate the target's input orthographic representation which, by spreading activation to the lexical phonological level, facilitates the production of the target name. This account appeals to different mechanisms for explaining the effect of phonological facilitation than the nonlexical account discussed above. Crucially, however, both accounts make identical predictions with respect to the asymmetry we found in the identity effects.

FIG. 3. (A) The hypothesis that two sources of activation are present in the identity effect observed with same-language pairs [the Catalan example *taula-taula* (*table*)]: one source is lexical, the other is nonlexical. (B) The identity effect observed with different-language pairs [e.g., the Catalan-Spanish pair *taula-mesa* (*table*)]. Activation is indicated by arrows.



explored the time course of the two types of identity effects. The time course of a distractor's effect can be studied by varying the time at which the word distractor is shown relative to the picture: the distractor can appear earlier than the picture (negative SOA) or can be delayed relative to the picture (positive SOA). In the present context, the time course of the semantic and the phonological effects are of particular interest. A temporal asymmetry between these two effects is typically reported in monolingual studies in which distractors were *visually* presented: while the semantic effect is observed at negative or 0 SOAs, the phonological effect arises at negative, 0, or positive SOAs (Damian & Martin, 1999; Glaser & Döngeloff, 1984; Glaser & Glaser, 1989; Starreveld & La Heij, 1996). This asymmetry is generally interpreted to mean that distinct levels of processing are involved in the semantic and the phonological effects (e.g., Schriefers et al., 1990). The semantic effect is assumed to arise at the level of lexical selection, whereas the phonological effect supposedly arises at the level where a word's phonological segments are retrieved. Consistent with this interpretation, and on the basis of our hypothesis about the identity effects in the same- and different-language conditions, we make the following predictions about the time course of the identity effects for the two conditions: If the same-language identity effect is a combination of two facilitatory effects (one lexical, the other phonological), such an effect should be observed *both* at negative and positive SOAs (Starreveld & La Heij, 1996). This prediction is based on the assumption that the same-language identity effect is supported by both lexical and phonological mechanisms. However, because the different-language identity effect is *only* supported by lexical mechanisms, such an effect should be observed only at negative or 0 SOAs. Furthermore, since the phonological effect arises at all SOAs, the same-language identity effect should be larger than the different-language identity effects regardless of SOA.

Method

Participants. Nineteen Catalan–Spanish bilingual speakers took part in the experiment.

Materials and procedure. We selected 16 experimental pictures and 16 filler pictures from the set of stimuli used in Experiment 2 (see Appendix 2d). Each experimental picture was matched with four word distractors: its Catalan name, its Spanish name, and a Catalan and a Spanish unrelated word. Unrelated words were noncognate Spanish–Catalan translations. Identical and unrelated distractors were of similar frequency [mean: identical = 32, unrelated = 18; $t(15) = 1.1$, n.s.] and length (mean number of letters = 6.0, range = 3–10; all F s were not significant). Two distractors had language-specific orthographic markings. Three SOA values were used: at -200 SOA the word preceded the appearance of the picture by 200 ms; at 0 SOA pictures and words were presented simultaneously; and at $+200$ SOA the picture preceded the appearance of the word by 200 ms. Across SOAs, the exposure duration of pictures and words remained constant (400 ms). Consequently, the amount of time in which pictures and words appeared simultaneously varied across SOAs—it was -200 ms for -200 and $+200$ SOAs and 400 ms for 0 SOA. Stimuli in the various experimental conditions were evenly distributed across 12 blocks of 32 trials each. The order of presentation of blocks was varied across subjects. In the training block, participants were presented with stimuli shown at all SOAs. With the exception of SOA variation, the procedure used in Experiment 4 was identical to that of Experiment 1. The experiment lasted about 40 min.

Results and Discussion

Table 6 shows mean response latencies and error rates for each experimental condition. A 2 (Distractor Language) $\times$ 2 (Target–Distractor Relation) $\times$ 3 (SOA) ANOVA was conducted. All main effects were significant: (a) response latencies were faster for distractors printed in Catalan than in Spanish [612 vs 647 ms; $F(1,18) = 50.7$, $MS_e = 1397.0$, $p < .001$; $F(1,15) = 21.7$, $MS_e = 2855$, $p < .001$]; (b) responses were faster for identical than unrelated distractors [600 vs 659 ms; $F(1,18) = 146.0$, $MS_e = 1351.2$, $p < .001$; $F(1,15) = 82.54$, $MS_e = 2046.3$, $p < .001$]; and (c) re-

TABLE 6

Mean Reaction Time, Standard Deviation (SD), and Error Percentage (E%) Across SOA's for Experiment 4

SOA	Language													
	Catalan							Spanish						
	Identical			Unrelated			Diff. ^a	Identical			Unrelated			Diff.
	RT	SD	E%	RT	SD	E%		RT	SD	E%	RT	SD	E%	
−200	523	57	(1.1)	649	51	(0.8)	+126	595	55	(1.0)	645	49	(1.0)	+50
0	565	43	(0.8)	706	64	(1.0)	+141	689	52	(1.0)	699	57	(0.9)	+10
+200	601	44	(1.1)	629	53	(1.1)	+28	629	46	(1.1)	627	47	(1.1)	−2

^a Diff stands for the difference between Identical and Unrelated distractors.

sponse latencies varied as a function of SOA [$F(1,2,36) = 69.9$, $MS_e = 1085.4$, $p < .001$; $F(2,30) = 55.4$, $MS_e = 1220.2$, $p < .0001$]. All two-way interactions were significant: (a) Distractor Language $\times$ Target–Distractor Relation [$F(1,18) = 75.1$, $MS_e = 1188.7$, $p < .001$; $F(2,15) = 33.2$, $MS_e = 2354.0$, $p < .001$]; (b) Distractor Language $\times$ SOA [$F(1,2,36) = 14.6$, $MS_e = 688.1$, $p < .001$; $F(2,30) = 9.3$, $MS_e = 899.0$, $p < .001$]; and (c) Target–Distractor Relation $\times$ SOA [$F(1,2,36) = 43.99$, $MS_e = 691.33$, $p < .001$; $F(2,30) = 16.7$, $MS_e = 1512.2$, $p < .001$]. Finally, the three-way interaction was also significant [$F(1,2,36) = 13.4$, $MS_e = 909.2$, $p < .001$; $F(2,30) = 10.2$, $MS_e = 1190.4$, $p < .001$]. Neuman–Keuls pairwise comparisons that contrasted identical to unrelated distractors were performed for the distractors of each language. For Catalan distractors, a significant difference ($p < .05$) between identical and unrelated distractors was found at all SOAs. A similar result was reported in monolingual studies which investigated the time course of the identity effect (see Glaser & Döngelhoff, 1984, Exp. 1; Glaser & Glaser, Exp. 6; Starreveld & La Heij, 1996, Exp. 1). In contrast, for Spanish distractors a significant difference appeared only at -200 SOA.

To summarize, two asymmetries were found in the facilitation effect for same- and different-language identity conditions. First, at all SOAs the effect was larger for same-language pairs (replicating Experiments 1 and 2). Second, the

identity effect varied across SOAs: for same-language distractors facilitation was demonstrated at each SOA (-200 , 0 , and $+200$) but for different-language distractors it was significant only at the negative SOA (although a trend was evident at 0 SOA). This pattern of results supports the hypothesis that part of the same-language identity effect may be attributed to phonological facilitation. Further support for this claim is sought in the next experiment.

EXPERIMENT 5: TRANSLATION-MEDIATED PHONOLOGICAL EFFECT

We have argued that part of the same-language identity effect involves phonological facilitation. We have further argued that this phonological effect can be attributed, at least in part, to nonlexical mechanisms which allow a distractor word to preactivate the target's phonological content. This explanation does not exclude that there may also be a lexical pathway through which the distractor activates the target's phonological segmental content. There is nothing in the account we have advanced that prevents activation from spreading from a lexical node to its corresponding phonological segments (see Fig. 3A). Our explanation only requires that if there is a lexical source to the phonological facilitation effect it is in addition to the nonlexical activation we have discussed here. The extent to which lexical factors contribute to the phonological facilitation effect is tested in Experiment 5. Consistent with our

TABLE 7

Examples of Target–Distractor Pairs for Experiment 5

Catalan name of the picture	Distractor language	Condition	
		Form related	Unrelated
<i>Baldufa</i> (spinning top)	Catalan	Baralla (fight)	Forquilla (fork)
		Translation related	Unrelated
	Spanish	Pelea (fight)	Tenedor (fork)

explanation of the identity effects, we expect to find that if there is a lexical contribution, it cannot account entirely for the effect of phonological facilitation.

In Experiment 5 we introduced a new type of target–distractor pair: Spanish distractors that once translated in Catalan are phonologically similar to Catalan targets (see Table 7). An example is the target–distractor pair *baldufa*–*pelea* (spinning top–fight). The Spanish distractor *pelea* is translated by the Catalan word *baralla*, which is phonologically similar to the Catalan target *baldufa*. The rationale for using this type of stimuli is the following (see Fig. 4): Consistent with the data reported in the preceding experiments, it is reasonable to assume that the Spanish distractor *pelea* activates the lexical form of its Catalan translation ‘baralla.’ If the lexical form ‘baralla’ activates its segmental features, some features of the target “baldufa” would also be activated and, therefore, target naming should be easier. As predicted by our account of the identity effects, however, a further result must also be observed: phonological facilitation with *baralla* (the Catalan translation of *pelea*). This is because *baralla* activates the segments of the target *baldufa* directly by means of nonlexical, orthography-to-phonology conversion processes. To test this prediction we included phonologically related Catalan distractors (like *baralla*). Importantly, evidence of facilitation with these words assures that their

Spanish translation (i.e., words like *pelea*) would also facilitate if there were any phonological facilitation through “translation.”

Method

Participants. Twenty Catalan–Spanish bilingual speakers took part in the experiment.

Materials and procedure. Twenty-three pictures were selected that were not cognates. Each picture was paired with four word distractors: two Catalan words (one phonologically related, the other unrelated) and their Spanish translations (both unrelated). The Catalan and Spanish words paired with a given picture were not cognates (see Appendix 2e). Phonologically related distractors shared the initial two or three segments with the target name. Related and unrelated distractors were matched for frequency [mean: related = 22, unrelated = 22; $t(22) < 1$] and length (mean number of letters = 5.9, range = 3–10; $F_s < 1$). Only one word had language-specific orthographic markings [*pañuelo* (Spanish, handkerchief)]. Twenty-three filler pictures, paired with four unrelated distractors (two Catalan words and two Spanish words), were also used. Stimuli were presented in four blocks of 46 trials each. The procedure was identical to that of Experiment 1. The entire experimental session lasted approximately 30 min.

Results and Discussion

As can be seen in Table 8, although a sizable phonological effect of 44 ms was found with Catalan related words, there is no evidence of phonological facilitation “through translation.” These observations were confirmed by the results of an analysis in which Distractor Type (related vs unrelated) and Distractor Language (Catalan vs Spanish) were examined. The main effect of target–distractor relatedness was significant [$F(1,19) = 12.5$, $MS_e = 817.8$, $p < .002$; $F(1,22) = 10.1$, $MS_e = 1149.3$, $p < .004$], but not that of distractor language ($F_s < 1$). Importantly, the interaction between these two variables was significant in the subject analysis [$F(1,19) = 17.4$, $MS_e = 487.1$, $p < .001$], although it only approached significance in the item analysis [$F(1,22) = 3.9$, $MS_e = 2592.2$,

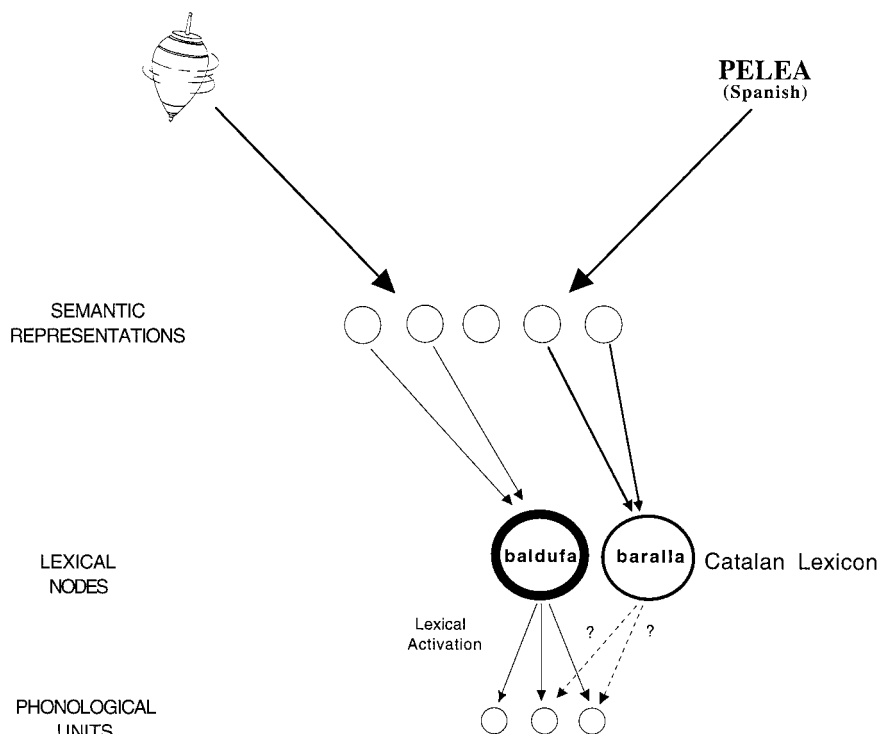


FIG. 4. Phonological effect “through translation.” The Catalan–Spanish pair *baldufa* (spinning top)–*pelea* (fight) is illustrated, in which the distractor’s Catalan translation (*baralla*) is phonologically related to the Catalan target. Whether the Spanish distractor *pelea* facilitates the naming of the Catalan target *baldufa* depends on whether the Catalan lexical form *baralla* (fight) can activate its phonemes. In the figure such activation is indicated by the broken line connecting the lexical node *baralla* to the phonological segments.

$p = .058$]. Pairwise comparisons (Newman–Keuls) revealed a significant difference between related and unrelated targets only for Catalan distractors.⁵ Thus, we can conclude that a distractor’s lexical node does not activate its corresponding segmental features (or if it does so the activation is too weak to lead to facilitation effects). This conclusion has three major implications.

First, phonological facilitation cannot be due to the fact that the distractor’s lexical node activates the phonological features of the target word. By further implication, it can be concluded that nonlexical mechanisms are responsible for the phonological effect. Second, the

⁵ In a pilot experiment in which 20 Catalan–Spanish speakers were tested, an identical null result was replicated with the same materials. The latter result confirms the reliability of the negative finding with this type of stimuli.

results of Experiment 5 are consistent with our account of the identity effects. This account assumes that phonological facilitation is due in large part to nonlexical mechanisms. Finally, the results of Experiment 5 have further implications for monolingual studies of the picture–word interference effect. The observations that (a) distractor lexical nodes do not seem to activate their segmental features and (b) distractor segmental features are activated (almost entirely) via sublexical mechanisms can be extended to the phenomenon of picture–word interference reported in monolingual studies. A detailed discussion of these conclusions and their relevance for other findings obtained in monolingual studies of speech production (e.g., Jescheniak & Schriefers, 1998; Levelt et al., 1991; Peterson & Savoy, 1998) is deferred until the General Discussion.

TABLE 8

Mean Reaction Time (RT), Standard Deviation (SD), and Error Percentage for Experiment 5

Distractor	Language					
	Catalan			Spanish		
	RT	SD	Error %	RT	SD	Error %
Related	751	68	3.4	778	85	2.4
Unrelated	795	73	4.5	780	74	4.1
Phonological effect (unrelated–related)	+44			–2		

EXPERIMENT 6: DIFFERENT-LANGUAGE
PHONOLOGICAL EFFECT

The results of the preceding experiments suggest that the effect of facilitation observed with phonologically related pairs is due at least in part to *sublexical* orthography-to-phonology conversion processes. The fact that evidence of phonological facilitation can be found with non-words is in agreement with this claim. The aim of Experiment 6 was to provide additional evidence in support of this claim by investigating the effects of cross-language, phonologically related target–distractors. For this purpose we used Catalan–Spanish distractor pairs like *baldufa* (spinning top)–*barro* (mud) (see Table 9). Since the target and the distractor of these pairs have different meanings, the distractor does not activate the target’s semantics nor does it affect the selection of the target’s lexical node. Furthermore, we know from the results of Experiment 5 that the distractor’s lexical representation does not activate the target’s segmental features. Therefore, the only way in which *barro* could facilitate the production of *baldufa* would be through nonlexical phonological facilitation. Such an effect would be consistent with the facilitation demonstrated with related nonword distractors in monolingual studies.

Another aim of Experiment 6 is to explore further the failure to obtain phonological facilitation “through translation” in Experiment 5. For this purpose we used Catalan distractors that when translated in Spanish were phonologically similar to the Catalan target (see Table 9). An example is the target–distractor pair *baldufa*

(Catalan, spinning top)–*fang* (Catalan, mud): the distractor is translated by the Spanish word *barro* (mud), which is similar to the Catalan target *baldufa*. By the same logic described in Experiment 5, if the distractor’s lexical representation can activate its phonological segmental content, we should observe facilitation “through translation.” Furthermore, evidence of phonological facilitation with *barro* would assure that *fang* should also facilitate if there were facilitation “through translation.”

Method

Participants. Twenty-two Catalan–Spanish bilingual speakers took part in the experiment.

Materials and procedure. Eighteen pictures with Catalan and Spanish noncognate names were selected. Each picture was matched with

TABLE 9
Examples of Target–Distractor Pairs for Experiment 6

Catalan name of the picture	Distractor language	Condition	
		Translation related	Unrelated
<i>Baldufa</i> (spinning top)	Catalan	Fang (mud)	Oli (oil)
		Form related	Unrelated
	Spanish	Barro (mud)	Aceite (oil)

TABLE 10

Mean Reaction Time (RT), Standard Deviation (SD), and Error Percentage for Experiment 6

Distractor	Language					
	Catalan			Spanish		
	RT	SD	Error %	RT	SD	Error %
Related	757	65	3.5	734	69	4.2
Unrelated	762	67	4.5	761	68	3.3
Phonological effect (unrelated–related)	+5			+27		

four word distractors: a phonologically related Spanish word, an unrelated Spanish word, and their Catalan translations (see examples in Table 9; the stimuli are reported in Appendix 2f). Since the orthographic systems of Catalan and Spanish are almost identical (see description in Appendix 1), Catalan–Spanish related words were both phonologically and orthographically similar. Related target–distractor nouns had the initial two or three segments in common. Catalan distractors were not phonologically related to the picture names. Catalan–Spanish translations were not cognates. Related and unrelated Spanish distractors were matched for frequency [mean: related = 27, unrelated = 28; $t(17) < 1$]. Two Spanish words had language-specific orthographic markings [*pañuelo* (handkerchief) and *otoño* (fall)]. The four groups of distractors were of similar length (mean number of letters = 5.6, range 3–10; $F_s < 1$). An additional set of 18 pictures (fillers) was selected and paired with four unrelated words (two in Catalan and two in Spanish). Stimuli were distributed across four blocks of 36 trials each. The experimental procedure was identical to that of Experiment 1.

Results and Discussion

Table 10 displays mean response latencies and error rates across the various groups of distractors tested in Experiment 6. The effects of Target Type and Target Language (Catalan vs Spanish) were both examined. Responses tended to be faster for related than unrelated distractors [745 vs 761 ms; $F(1,21) = 5.7$,

$MS_e = 942.4$, $p < .03$; but $F(1,17) = 12.1$, $MS_e = 1416.5$, $p = .08$]. The effect of language was not significant [$F(1,21) = 3.9$, $MS_e = 816.4$; n.s.; $F(1,17) = 2.2$, $MS_e = 1314.3$, n.s.]. The interaction between these variables was significant only in the subject analysis [$F(1,21) = 4.1$, $MS_e = 682.0$, $p < .05$; $F(1,21) = 1.6$, $MS_e = 1746.4$, n.s.]. Post hoc comparisons (Newman–Keuls) revealed a significant difference between Spanish related and unrelated distractors. There was no significant difference between the two groups of Catalan distractors.

Experiment 6 revealed that a sizable phonological effect can be demonstrated with target–distractor pairs of different languages. Furthermore, as in Experiment 5, we failed to provide evidence of phonological facilitation “through translation.” Namely, the fact that the distractor once translated is phonologically related to the target does not affect naming latencies. In brief, the results of Experiment 6 support the predictions derived from the hypothesis that the effect of phonological facilitation (or a substantial part of this effect) has a nonlexical basis.

EXPERIMENT 7: THE TIME COURSE OF THE TRANSLATION-MEDIATED PHONOLOGICAL EFFECT

It could be argued that our failure to observe phonological facilitation “through translation” reflects the SOA values used in Experiments 5 and 6. Perhaps it takes some time for a word to fully activate its translation. Thus, for example, when the Catalan lexical node ‘baralla’ receives

TABLE 11

Mean Reaction Time, Standard Deviation (SD), and Error Percentage (E%) across SOA's for Experiment 7

SOA	Language												
	Catalan							Spanish					
	Form related			Unrelated			Diff. ^a	Translation related			Unrelated		
	RT	SD	E%	RT	SD	E%		RT	SD	E%	RT	SD	E%
-200	716	74	(0.6)	743	65	(0.8)	+27	722	80	(0.7)	721	73	(0.6)
0	742	59	(0.9)	800	76	(1.2)	+58	781	82	(0.9)	783	67	(0.7)
+200	716	69	(0.8)	731	82	(0.8)	+15	739	71	(0.8)	735	77	(0.7)

^a Diff stands for the difference between Form Related and Unrelated distractors.

activation from its Spanish translation *pelea* it might be too late to affect the retrieval of the phonological features of the target ‘*baldufa*.’ This hypothesis makes a straightforward prediction: by anticipating the presentation of the distractor we allow time for translation to take place, and thus we should find an effect of phonological facilitation with Spanish–Catalan pairs like *pelea*–*baldufa*. To test this prediction, pairs like *pelea*–*baldufa* were presented at –200, 0, and +200 SOAs.

Method

Participants. Seventeen Catalan–Spanish bilingual speakers took part in the experiment.

Materials and procedure. The material was selected following the criteria described in Experiment 5. We selected 21 pictures with Spanish–Catalan noncognate names. Pictures were paired with four word distractors: two Catalan words (one phonologically related, the other unrelated) and their noncognate Spanish translations (the stimuli are listed in Appendix 2g). Phonologically related distractors shared the initial two or three segments with the target name. Related and unrelated distractors were matched for frequency and length ($F_s < 1$). One distractor had language-specific orthographic markings [*otoño* (Spanish, fall)]. We also used 21 filler pictures that were paired with four unrelated distractors (two Catalan words and two Spanish words). All pictures were presented at SOA –200, 0, and +200. Stimuli of

the various experimental conditions were evenly distributed across 12 blocks of 42 trials each. The procedure was the same as in Experiment 4.

Results and Discussion

Table 11 shows the response latencies and error rates as a function of distractor language, phonological relatedness, and SOA. The effects of same- and different-language distractors were examined in separate analyses. With same-language distractors, responses were faster for related than unrelated words [$F(1,16) = 36.5$, $MS_e = 768.7$, $p < .001$; $F(1,20) = 15.2$, $MS_e = 2357$, $p < .001$]. This discrepancy varied across SOAs, as indicated by a significant interaction between Type of Distractor and SOA [$F(1,32) = 3.5$, $MS_e = 1156.7$, $p < .05$; though $F(2,40) = 1.9$, $MS_e = 2667.8$, n.s.]. In contrast, no trace of phonological facilitation was found with different-language distractors across SOAs ($F_s < 1$). The latter result replicates and further extends the finding of Experiments 5 and 6. More importantly, this result disconfirms the hypothesis that the lack of phonological facilitation “through translation” is due to the time required to translate the word distractor.

GENERAL DISCUSSION

We have reported a series of bilingual picture–word interference experiments in which

Catalan–Spanish speakers named pictures in Catalan. The following results were obtained.

Identical Distractors

Facilitation was found for both same- and different-language identity distractors. These effects are robust, having been replicated in several experiments and in a variety of conditions (e.g., blocked vs mixed language naming). However, two differences were observed for same- and different-language pairs. The facilitation effect was *larger* for same-language pairs and it was found across *all* SOAs (–200, 0, and +200) for same-language pairs but only at negative (–200) and 0 SOAs for different-language pairs.⁶

Semantically Related Distractors

Semantic interference was obtained for both Catalan and Spanish distractors. The size of the semantic effect was the *same* in the same- and different-language conditions. Similar results have been reported in other investigations in which highly proficient bilinguals were tested (e.g., Ehry & Ryan, 1980; Goodman, et al. 1985; Mägiste, 1984; Potter, et al., 1984; Smith and Kisner, 1982).

Phonologically Related Distractors

Phonologically similar distractors facilitated picture naming *regardless* of their language. However, we failed to observe phonological facilitation “*through translation*.” That is, Spanish distractors that translate into phonologically similar words to the target response did not produce facilitation [e.g., the Spanish distractor *pelea* (fight), which is translated as *baralla* in Catalan, did not facilitate the production of the Catalan target *baldufa* (spinning top)].

The results obtained with identical, semantically related, and phonologically related distractors can be accounted for by a model of lexical access that makes the following assumptions: (a) the semantic system sends activation in parallel and to equal extents to the lexical entries in

the two lexicons of a bilingual; (b) *only* the lexical nodes in the lexicon which is programmed for response are considered for selection; and (c) there are *nonlexical* mechanisms that allow a written word to activate its phonological segments. It should be noted that with the exception of the mechanisms that allow for language-specific selection of lexical representations, all the other mechanisms are part of the monolingual production system.

Within this type of model, the pattern of results we have reported can be explained as follows: The facilitation obtained with cross-language identical pairs such as *taula* (Catalan, table)–*mesa* (Spanish, table) reflects the fact the Spanish distractor activates not only its corresponding lexical representation ‘*mesa*’ but also its Catalan counterpart ‘*taula*.’ Because only Catalan lexical forms are considered for selection, the lexical representation ‘*mesa*,’ though highly activated by the word distractor (and the picture of *table*), cannot interfere with the selection of the Catalan target “*taula*.” Furthermore, since the target “*taula*” receives extra activation from *mesa*, its production is facilitated relative to a neutral distractor.

The semantic interference observed across languages can be explained in a similar manner. Consider the case in which the picture *table* (in Catalan “*taula*”) is shown with the related Spanish word *silla* (chair). The distractor activates both the Spanish form ‘*silla*’ and its Catalan translation ‘*cadira*.’ However, because selection is assumed to be restricted to the lexical nodes within the Catalan lexicon, ‘*cadira*’ instead of ‘*silla*’ will interfere with the retrieval of the Catalan target “*taula*.” Furthermore, since the semantic representations of *silla* activates equally strongly ‘*silla*’ and ‘*cadira*,’ the interference produced by the Spanish distractor *silla* is indistinguishable from that of its Catalan translation *cadira*.

The finding that semantic interference is not affected by distractor language is problematic for accounts that postulate asymmetrical activation of the two lexicons. For example, Poulisse and Bongaerts (1994; see also Poulisse, 1997) proposed that lexical nodes (lemmas) are activated both by the semantic and by a “language

⁶ Evidence of facilitation with different-language identity distractors is also obtained when speakers named pictures in their nondominant language (Costa & Caramazza, submitted).

component.” The latter specifies, for each instance, the language that has to be selected (e.g., if the target language is Catalan, the forms of the Catalan lexicon are activated; for a similar proposal see also Grainger & Dijkstra, 1992). By this account, we should have observed larger semantic interference with Catalan distractors. Our results are similarly problematic for Green’s (1986) hypothesis that words in the nonresponse language are inhibited.

Finally, to account for the larger facilitation effect obtained with the same- than the different-language identity pairs, we must consider the contribution of a nonlexical (or nonsemantic) orthography-to-phonology conversion mechanism in the picture–word interference task. This mechanism has been invoked in order to account for the phonological facilitation observed with related pairs (e.g., *table–tailor*; *chair–chain*) in monolingual experiments (e.g., Roelofs et al., 1996). When the word distractor coincides with the target name, all the segments of the target are activated by the distractor. As a consequence of such activation, target naming should be easier. The same type of activation cannot be obtained when the distractor is the translation of the target name and is not a cognate word [as in *taula* (Catalan, table)–*mesa* (Spanish, table)]. Since the latter distractors do not share phonological content with the target there is no activation of the target’s phonological content through nonlexical mechanisms. Therefore, the fact that same-language identical distractors (e.g., *taula–taula*) but not different-language identical distractors (e.g., *taula–mesa*) can activate the phonological content of the target explains the asymmetry that we observed between same- and different-language conditions.

Although the results reported here have implications primarily for models of bilingual lexical access, they are also relevant for determining the causes of the phonological facilitation effect observed in the picture–word naming task. As noted in the Introduction, there are at least two ways in which word distractors can activate the phonological segments of the target word. One is through nonlexical processes that convert the orthographic stimulus into a phono-

logical representation. That mechanisms of this sort are involved in phonological facilitation is suggested by the fact that this effect can be found with nonword distractors. It is also possible, however, that distractor words activate their phonological segmental content through their lexical forms. If this were the case, however, we should have found evidence of phonological facilitation “through translation.” Thus, according to our data, it seems reasonable to conclude that the phenomenon of phonological facilitation is (mostly) due to sublexical mechanisms that allow the direct activation of the target’s phonological segments.

This conclusion is in agreement with results recently reported by Peterson and Savoy (1998). In their study, participants saw a picture (e.g., *couch*) followed at varying interval by either a cue signaling that the pictured was to be named or by word which was to be read aloud. Some target words were semantically related (*bed*) or phonologically related (*count*) to the picture (*couch*). A priming effect was found with both semantically and phonologically related words. Interestingly, however, no evidence of priming was observed with words phonologically similar to an associate of the picture—for example the picture *couch* did not prime the word *bet* (phonologically related to *bed*). Thus, despite the fact that *couch* activates ‘bed,’ as demonstrated by the results of semantic priming, it does not seem that ‘bed’ activates its phonological features. This result suggests that lexical forms that are not selected activate their phonological features only very weakly (if at all). Other data reported by Peterson and Savoy require that we modulate this conclusion. They found that the picture *couch* primed the reading of *soda*—a word phonologically related to *sofa*, a near-synonym of *couch* (for converging evidence see Jescheniak and Schriefers, 1998). As suggested by Peterson and Savoy (1998), the contrasting results obtained with words like *bet* and *soda* might reflect differences in the activation levels of the related lexical nodes ‘bed’ and ‘sofa.’ That is, when the picture *couch* is presented the near-synonym ‘sofa’ is far more activated than its related word ‘bed’ because it is a possible response for the picture. This implies

that the activation of phonological features by distractor items is generally very weak and it is only in very special cases (near synonyms, for example) that there is enough activation of the lexical node to spread to the segmental level.

The language abilities of bilinguals can vary extensively—from individuals who are native speakers of two languages to those who learn a second language much later as part of their school curriculum. The model we propose is not designed to account for lexical access in all the various groups of bilingual speakers. Its scope is more limited: it is aimed at describing lexical access in highly proficient, balanced bilingual speakers like the participants in our experiments. However, with additional assumptions the model can be extended to lexical access by less proficient bilinguals. For example, as proposed by Kroll and Stewart (1994), it could be that the semantic system activates the native language (L1) lexicon more intensely than the second language (L2) lexicon. Alternatively, it could be proposed, as Poulisse and Bongaerts (1994) have done, that the advantage for selecting L1 forms reflects word frequency—high-familiarity words (most probably from L1) are more easily accessed than low-familiarity words (most probably from L2). As proficiency and familiarity increase, the selection of L2 forms improves.

One example of how the speech production of bilinguals may depend on their proficiency is found in the recent study conducted by Hermans, Bongaerts, de Bot and Schreuder (1998). In this study, Dutch–English bilinguals participated in a series of picture–word interference tasks in which they named pictures in English (their nondominant language). Of particular interest in this context is the condition in which distractors were Dutch words [e.g., *berm* (verge)] that were phonologically related to the Dutch translation [e.g., *berg* (mountain)] of the English target (“mountain”). These distractors produced interference—e.g., it took longer to respond to “mountain” with the Dutch distractor *berm* (verge) than with the unrelated distractor *kaars* (candle). Hermans et al. interpreted this result as evidence that the nonresponse language interferes with the selection of the target lexical node in the response language. This in-

terpretation is in contradiction with the model proposed here. However, there are at least two major differences between Hermans et al.’s study and the experiments reported here that may explain the contrasting results. First, the response language in the two studies was different. While in our experiments participants were required to name the pictures in L1, in Herman et al.’s experiments the response language was L2. Second, the Catalan–Spanish bilinguals tested in our experiment were much more proficient bilinguals than the Dutch–English bilinguals tested by Hermans et al. The functioning of the language-specific selection mechanism may depend on both the bilingual’s proficiency and the response language. It is reasonable to assume that the more proficient a bilingual is, the easier it is to restrict lexical selection to only one language. In addition, if language specific selection is not completely functional, it is more likely to affect participants’ performance when naming in L2. Further research is needed to determine how the response language and the bilinguals’ language proficiency may affect the language-specific selection mechanism.

In conclusion, the results reported in this article support the hypothesis of language-specific selection in bilingual lexical access. A precise account of this mechanism needs to explain how the lexical systems are distinguished and what information governs language selection. We hope that future investigations will clarify these important issues.

APPENDIX 1

Catalan and Spanish are both Romance languages. Nouns of both languages are overtly marked for gender (masculine vs feminine) and number. These languages have different vowel repertoires: Spanish has a relatively small inventory of five vowels (/a/, /e/, /i/, /o/, /u/), which can be realized both in stressed and unstressed positions; Catalan has eight vowels, (/a/, /e/, /ɛ/, /i/, /ɔ/, /o/, /u/, and /ə/), with vowel reduction in unstressed position (/a/, /e/, and /ɛ/ are reduced to schwa /ə/ and /o/ are reduced to /u/). The consonant repertoire is also different: the consonants /θ/, /v/, and /x/ can be found only in Spanish, and the consonants /ʃ/, /ʒ/, /w/, /j/, and /z/ can be found only in Catalan. Differences are also present in the phonotactic constraints of each language. For example, in Catalan several combinations of consonants clusters can be found in syllable codas (*fals*, *mans*, *cars*,

mots), while in Spanish only one cluster is permissible (construir). In the orthographic system, only Spanish has the grapheme n and only Catalan has the graphemes i and ç. The phoneme /tʃ/ is realized by the graphemes ch and tx in Spanish and Catalan, respectively. Finally, in Catalan two signs are used to indicate stress (as in e and è) but only one in Spanish (é).

In Catalonia, Catalan and Spanish are both official languages. In many families, both languages are spoken. In kindergarten (ages 4–5), special Catalan programs are offered to children from monolingual Spanish families. The current education system requires that at the end of the primary school (year 11/12), children are able to read, write, speak, and understand both Catalan and Spanish. In high school, some classes are taught in Catalan and others in Spanish. At the university, classes and tests can be in either

language—quite often half of the test is in Catalan, the other half in Spanish. Radio and television programs are broadcast in Catalan and in Spanish. Furthermore, some newspapers contain articles written in Catalan and in Spanish.

Our group of participants was relatively homogenous with respect of bilingual proficiency. All the participants in our experiments reported speaking Catalan with at least one of their parents. They were native of Catalonia and attended to bilingual schools for at least 13 years. All the participants passed the Catalan–Spanish language proficiency exam that is required for enrollment at the university. In order to pass this exam students must be very proficient in all aspects of the two languages (vocabulary, grammar, etc.) in written and verbal comprehension and production. Participants reported to use both languages in their daily life.

APPENDIX 2A

Stimuli for Experiment 1

Picture	Distractors/language			
	Identical		Control	
	Catalan	Spanish	Catalan	Spanish
Anec (duck)	Anec	Pato	Enciam	Lechuga (lettuce)
Barret (hat)	Barret	Sombrero	Clau	Llave (key)
Alberginia (eggplant)	Alberginia	Berenjena	Queixal	Muela (tooth)
Coixi (pillow)	Coixi	Almohada	Nina	Muñeca (doll)
Colom (dove)	Colom	Paloma	Llar	Hogar (home)
Dona (woman)	Dona	Mujer	Ull	Ojo (eye)
Finestra (window)	Finestra	Ventana	Por	Miedo (fear)
Fulla (leaf)	Fulla	Hoja	Sopar	Cena (dinner)
Gelat (ice cream)	Gelat	Helado	Branca	Rama (branch)
Papallona (butterfly)	Papallona	Mariposa	Matalas	Colchon (mattress)
Gos (dog)	Gos	Perro	Fusta	Madera (wood)
Poma (apple)	Poma	Manzana	Escacs	Ajedrez (chess)
Pastanaga (carrot)	Pastanaga	Zanahoria	Genoll	Rodilla (knee)
Ganivet (knife)	Ganivet	Cuchillo	Mocador	Pañuelo (handkerchief)
Mitjo (sock)	Mitjo	Calcetin	Formatge	Queso (cheese)
Granota (frog)	Granota	Rana	Esquina	Cantonada (corner)
Porc (pig)	Porc	Cerdo	Aixeta	Grifo (faucet)
Taula (table)	Taula	Mesa	Pernil	Jamon (ham)

APPENDIX 2B

Stimuli for Experiment 2

Picture	Distractors/language					
	Identical		Semantically related		Control	
	Catalan	Spanish	Catalan	Spanish	Catalan	Spanish
Mitjo (sock)	Mitjo	Calcetin	Guant	Guante (glove)	Fresc	Fresco (cool)
Forquilla (fork)	Forquilla	Tenedor	Plat	Plato (plate)	Part	Parte (part)
Ull (eye)	Ull	Ojo	Brac	Brazo (arm)	Grup	Grupo (group)
Colom (dove)	Colom	Paloma	Gavina	Gaviota (seagull)	Vestit	Vestido (dress)
Alberginia (eggplant)	Alberginia	Berenjena	Carabasso	Calabacin (zucchini)	Sabatilla	Zapatilla (slipper)
Llamp (lightning)	Llamp	Rayo	Vent	Viento (wind)	Gra	Grano (pimple)
Esquirol (squirrel)	Esquirol	Ardilla	Ratoli	Raton (mouse)	Calaix	Cajon (draw)
Mongeta (bean)	Mongeta	Judia	Llentia	Lenteja (lentil)	Bitllet	Billete (check)
Samarreta (shirt)	Samarreta	Camiseta	Abrig	Abrigo (coat)	Rebut	Recibo (receipt)
Gelat (ice cream)	Gelat	Helado	Pastis	Pastel (cake)	Pilota	Pelota (ball)
Gos (dog)	Gos	Perro	Llop	Lobo (wolf)	Marc	Marco (frame)
Rentadora (washing machine)	Rentadora	Lavadora	Cuina	Cocina (stove)	Conill	Conejo (rabbit)
Barret (hat)	Barret	Sombrero	Casc	Casco (helmet)	Colze	Codo (elbow)
Cuc (worm)	Cuc	Gusano	Pussa	Pulga (flea)	Molsa	Musgo (moss)
Poma (apple)	Poma	Manzana	Llimona	Limon (lemon)	Llista	Lista (list)
Papallona (butterfly)	Papallona	Mariposa	Abella	Abeja (bee)	Jutge	Juez (judge)
Formatge (cheese)	Formatge	Queso	Mantega	Mantequilla (butter)	Perill	Peligro (peril)
Taula (table)	Taula	Mesa	Armari	Armario (closet)	Palau	Palacio (palace)
Ganivet (knife)	Ganivet	Cuchillo	Navalla	Navaja (jackknife)	Llengua	Lengua (tongue)
Destral (axe)	Destral	Hacha	Serra	Sierra (saw)	Metge	Medico (doctor)
Porc (pig)	Porc	Cerdo	Ovella	Oveja (sheep)	Ovul	Ovulo (ovum)
Os (bone)	Os	Hueso	Cor	Corazon (heart)	Cec	Ciego (blindman)

APPENDIX 2C

Stimuli for Experiment 3

Picture	Distractors/language			
	Semantically related		Control	
	Catalan	Spanish	Spanish	Catalan
Coixi (pillow)	Llensol	Sabana (blanket)	Papallona	Mariposa (butterfly)
Forquilla (fork)	Ganivet	Cuchillo (knife)	Onada	Ola (wave)
Mongeta (bean)	Pesol	Guisante (pea)	Pardal	Gorrión (sparrow)
Taula (table)	Cadira	Silla (chair)	Got	Vaso (glass)
Pastanaga (carrot)	Alberginia	Berenjena (eggplant)	Mirall	Espejo (mirror)
Fulla (leaf)	Branca	Rama (branch)	Presso	Carcel (jail)
Falç (sickle)	Destral	Hacha (axe)	Llavor	Semilla (seed)
Poma (manzana)	Pressec	Melocoton (peach)	Clauer	Llavero (key ring)
Llit (bed)	Matalas	Colchon (mattress)	Besso	Gemelo (twin)
Raim (grape)	Pruna	Ciruela (plum)	Guenyo	Bizzo (cross-eyed)
Mitjo (sock)	Esclop	Zueco (clog)	Estalvi	Ahorro (saving)
Colom (dove)	Mussol	Buho (owl)	Genoll	Rodilla (knee)
Esquirol (squirrel)	Guineu	Zorro (fox)	Amanida	Ensalada (salad)
Porc (pig)	Vedella	Ternera (calf)	Malalt	Enfermo (sick)
Anec (duck)	Oreneta	Golondrina (swallow)	Deixeble	Discipulo (disciple)
Estel (kite)	Baldufa	Peonza (spinning top)	Suro	Corcho (cork)
Formatge (cheese)	Pernil	Jamon (ham)	Cosi	Primo (cousin)
Pluja (rain)	Boira	Niebla (fog)	Galta	Mejilla (cheek)
Llesca (slice)	Entrepa	Bocadillo (sandwich)	Lloguer	Alquiler (rent)

APPENDIX 2D

Stimuli for Experiment 4

Picture	Distractors/language			
	Identical		Control	
	Catalan	Spanish	Spanish	Catalan
Barret (hat)	Barret	Sombrero	Clau	Llave (key)
Alberginia (eggplant)	Alberginia	Berenjena	Queixal	Muela (molar)
Coixi (pillow)	Coixi	Almohada	Nina	Muñeca (doll)
Cuc (worm)	Cuc	Gusano	Baralla	Pelea (fight)
Destral (axe)	Destral	Hacha	Estalvi	Ahorro (saving)
Esquirol (squirrel)	Esquirol	Ardilla	Tret	Rasgo (trait)
Fulla (leaf)	Fulla	Hoja	Sopar	Cena (dinner)
Gelat (ice cream)	Gelat	Helado	Colom	Paloma (dove)
Mongeta (bean)	Mongeta	Judia	Galleda	Cubo (bucket)
Papallona (butterfly)	Papallona	Mariposa	Matalas	Colchon (mattress)
Ull (eye)	Ull	Ojo	Estona	Rato (while)
Gos (dog)	Gos	Perro	Fusta	Madera (wood)
Poma (apple)	Poma	Manzana	Escacs	Ajedrez (chess)
Pastanaga (carrot)	Pastanaga	Zanahoria	Genoll	Rodilla (knee)
Formatge (cheese)	Formatge	Queso	Brau	Toro (bull)
Ganivet (knife)	Ganivet	Cuchillo	Mocador	Pañuelo (handkerchief)

APPENDIX 2E

Stimuli for Experiment 5

Picture	Distractors/language			
	Phonologically related		Control	
	Catalan	Spanish	Catalan	Spanish
Falç (sickle)	Fastig	Asco (disgust)	Mandra	Pereza (laziness)
Barret (hat)	Batec	Latido (beat)	Presso	Carcel (jail)
Galleda (bucket)	Ganivet	Cuchillo (knife)	Clau	Llave (key)
Pardal (sparrow)	Patir	Sufrir (to suffer)	Enfonsar	Hundir (to sink)
Forquilla (fork)	Forat	Agujero (hole)	Ampolla	Botella (bottle)
Esquirol (squirrel)	Escacs	Ajedrez (chess)	Baralla	Pelea (fight)
Papallona (butterfly)	Pastanaga	Zanahoria (carrot)	Matalas	Colchon (mattress)
Gelat (ice cream)	Genoll	Rodilla (knee)	Llensol	Sabana (blanket)
Coixi (pillow)	Colom	Paloma (dove)	Groc	Amarillo (yellow)
Capella (priest)	Cadira	Silla (chair)	Malalt	Enfermo (sick)
Blat (wheat)	Blau	Azul (blue)	Llar	Hogar (home)
Finestra (window)	Filferro	Alambre (wire)	Peto	Beso (kiss)
Vedella (calf)	Vermell	Rojo (red)	Xarop	Jarabe (syrup)
Fulla (leaf)	Fusta	Madera (wood)	Pluja	Lluvia (rain)
Raspall (brush)	Raim	Uva (greip)	Boig	Loco (crazy)
Mitjo (sock)	Mirall	Espejo (mirror)	Cosi	Primo (cousin)
Guineu (fox)	Guix	Tiza (chalk)	Fang	Barro (mug)
Got (glass)	Gos	Perro (dog)	Taula	Mesa (table)
Sargantana (lizard)	Samarreta	Camiseta (shirt)	Enciam	Lechuga (lettuce)
Civella (belt)	Cigro	Garbanzo (chickpea)	Estalvi	Ahorro (saving)
Cargol (snail)	Cantonada	Esquina (corner)	Foscor	Oscuridad (darkness)
Mongeta (bean)	Mocador	Pañuelo (handkerchief)	Clauer	Llavero (key ring)
Llit (bed)	Lligar	Atar (to tie)	Triar	Escoger (to choose)

APPENDIX 2F

Stimuli for Experiment 6

Picture	Distractors/language			
	Phonologically related		Control	
	Catalan	Spanish	Catalan	Spanish
Papallona (butterfly)	Mocador	Pañuelo (handkerchief)	Genoll	Rodilla (knee)
Raspall (brush)	Branca	Rama (branch)	Matalas	Colchon (mattress)
Mitjo (sock)	Por	Miedo (fear)	Forat	Agujero (hole)
Cuc (worm)	Galleda	Cubo (bucket)	Pluja	Lluvia (rain)
Pressec (peach)	Cosi	Primo (cousin)	Tardor	Otoño (fall)
Mussol (owl)	Queixal	Muela (tooth)	Soroll	Ruido (noise)
Raim (grape)	Cua	Rabo (tail)	Guix	Tiza (chalk)
Pastanaga (carrot)	Ocell	Pajaro (bird)	Clau	Llave (key)
Baldufa (spinning top)	Fang	Barro (mud)	Oli	Aceite (oil)
Cadira (chair)	Presso	Carcel (jail)	Enciam	Lechuga (lettuce)
Dona (woman)	Vora	Dobladillo (hem)	Baralla	Pelea (fight)
Gos (dog)	Cop	Golpe (punch)	Ull	Ojo (eye)
Bec (beak)	Peto	Beso (kiss)	Fastic	Asco (disgust)
Sargantana (lizard)	Llensol	Sabana (blanket)	Cantonada	Esquina (corner)
Pesol (pea)	Mandra	Pereza (laziness)	Coto	Algodon (cotton)
Blat (wheat)	Tou	Blando (soft)	Lloguer	Alquiler (rent)
Mirall (mirror)	Dimecres	Miercoles (Wednesday)	Forquilla	Tenedor (fork)
Granota (frog)	Aixeta	Grifo (faucet)	Samarreta	Camiseta (shirt)

APPENDIX 2G

Stimuli for Experiment 7

Picture	Distractors/language			
	Phonologically related		Control	
	Catalan	Spanish	Catalan	Spanish
Forquilla (fork)	Foscor	Oscuridad (darkness)	Soroll	Ruido (noise)
Falç (sickle)	Fastig	Asco (disgust)	Mandra	Pereza (laziness)
Taula (table)	Taca	Mancha (stain)	Genoll	Rodilla (knee)
Colom (dove)	Cosi	Primo (cousin)	Boig	Loco (crazy)
Papallona (butterfly)	Pastanaga	Zanahoria (carrot)	Matalas	Colchon (mattress)
Raspall (brush)	Raim	Uva (grape)	Pluja	Lluvia (rain)
Formatge (cheese)	Forat	Agujero (hole)	Tardor	Otoño (fall)
Llesca (slice)	Llensol	Sabana (blanket)	Blau	Azul (blue)
Baldufa (spinning top)	Baralla	Pelea (fight)	Queixal	Muela (tooth)
Esquirol (squirrel)	Escacs	Ajedrez (chess)	Boira	Niebla (fog)
Esclop (clog)	Estalvi	Ahorro (saving)	Oli	Aceite (oil)
Mitjo (sock)	Mirall	Espejo (mirror)	Entropa	Bocadillo (sandwich)
Llit (bed)	Lligar	Atar (to tie)	Menjar	Comer (to eat)
Poma (apple)	Por	Miedo (fear)	Guix	Tiza (chalk)
Cuc (worm)	Cuixa	Muslo (thigh)	Branca	Rama (branch)
Cadira (chair)	Cantonada	Esquina (corner)	Pesol	Guisante (pea)
Ganivet (knife)	Galleda	Cubo (bucket)	Parla	Habla (speech)
Pressec (peach)	Presso	Carcel (jail)	Estiu	Verano (summer)
Xarxa (net)	Xarop	Jarabe (syrup)	Enciam	Lechuga (lettuce)
Vedella (calf)	Vermell	Rojo (red)	Clau	Llave (key)
Fulla (leaf)	Fusta	Madera (wood)	Aixeta	Grifo (faucet)

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